FB(S³)R: Fractal S³ Cosmology — A Window-Anchored Gradient Account of the Hubble Tension

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Objective. We formulate a compact, simply connected, boundaryless three-sphere (S^3) cosmology with a discrete fractal embedding ("φ-ladder") in which the observed Hubble slope is a *windowed gradient* rather than a global expansion rate.
Postulate FB-R. Redshift is introduced via a covariant, coarse-grained transport law along null geodesics, $\mathrm{d}\ln\omega/\mathrm{d}\lambda = -\kappa\, p(V)\,(1/V)\,\mathrm{d}V/\mathrm{d}\lambda$; FRW matching in the homogeneous limit (with $p = 1/3$) fixes $\kappa = 1$. To render absolute z finite (e.g., to the CMB) while preserving differential relations, we regularize the observer-side divergence with a minimum coarse-graining angle $\chi_{\min} \sim \ell_{\min}/R$.
Observable (windowed) Hubble slope. Observables are finite-resolution convolutions of the infinitesimal gradient, yielding $H(\ell) = 3c\,p/\ell$ after a canonical window normalization that removes the near-observer $1/r$ divergence and recovers the linear Hubble law.
Two-regime prediction. Distinct rarefaction indices and windows, $(p_{\rm rec}, p_{\rm loc}) \approx (4/3, 1)$ with $\ell_{\rm eff}^{\rm rec} > \ell_{\rm eff}^{\rm loc}$, give $H_{\rm CMB} = 3c\,p_{\rm rec}/\ell_{\rm eff}^{\rm rec}$ and $H_{\rm late} = 3c\,p_{\rm loc}/\ell_{\rm eff}^{\rm loc}$, and the sharp *ratio law* $H_{\rm late}/H_{\rm CMB} = (p_{\rm loc}/p_{\rm rec})\,(\ell_{\rm eff}^{\rm rec}/\ell_{\rm eff}^{\rm loc})$. A fractional φ-step $\Delta n \simeq 0.78$ between CMB and late windows implies $\ell_{\rm eff}^{\rm rec}/\ell_{\rm eff}^{\rm loc} \simeq \varphi^{0.78} \simeq 1.45 \Rightarrow H_{\rm late}/H_{\rm CMB} \simeq 1.09$, matching the observed 8–9% split without tuning.
Consistency. Etherington reciprocity holds *exactly* ($\eta \equiv 1$) under geodesic photon flow and photon-number conservation. Early-time microphysics is unchanged.
Falsifiable signatures. (i) A distinctive redshift-drift curve with a shifted zero-crossing, (ii) a φ-correlated BAO phase offset $\mathcal{O}(10^{-2})$, (iii) window-consistent φ-step linking $H_{\rm CMB}$ and $H_{\rm late}$. We delineate scope precisely: the two-slope phenomenology arises without extra fluids; full global fits (CMB+BAO+SN+BBN) remain as future work.

I. INTRODUCTION

A robust split persists between the Hubble slope inferred near recombination ($\sim 67\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [1]) and that from late-time distance ladders ($\sim 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [2, 3]). Within ΛCDM a single scale-factor history must bridge both, so the split is framed as the "Hubble tension," often approached via additional components (e.g., early dark energy) and parameter tuning.

We pursue a different perspective. In FB(S^3)R the Universe has compact, simply connected spatial sections S^3. Cosmological redshift is treated not as Doppler recession in an "expanding meta-space" but as an *adiabatic density-gradient integral* along S^3 null geodesics. The observed "Hubble law" is, operationally, a *windowed gradient*: different observational windows (CMB vs. late-time candles/rulers) probe different coarse-graining scales and effective rarefaction indices, naturally yielding two slopes in the observed range—without tuned new sectors. We do not claim a solved global fit; our claim is the ratio law and its window anchoring.

II. GEOMETRY ON S^3 AND THE φ-LADDER

A. S^3 measure and geodesic volume

Let spatial slices be S^3 with curvature radius R and angular radial coordinate $\chi \in [0, \pi]$; geodesic distance $r = R\chi$. The 3-ball volume of angular radius χ is

$$V(\chi) = 2\pi R^3\big(\chi - \tfrac{1}{2}\sin 2\chi\big)\,, \tag{1}$$

$$\frac{\mathrm{d}V}{\mathrm{d}\chi} = 4\pi R^3 \sin^2\chi, \qquad \frac{\mathrm{d}V}{\mathrm{d}r} = 4\pi R^2 \sin^2\chi. \tag{2}$$

Define the dimensionless kernel

$$K(\chi) \equiv \frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}\chi} = \frac{2\sin^2\chi}{\chi - \sin\chi\cos\chi}, \qquad \frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}r} = \frac{K(\chi)}{R}. \tag{3}$$

Asymptotics: for $\chi \to 0$, $K(\chi) = \frac{3}{\chi} - \frac{2\chi}{5} + O(\chi^3)$, recovering $(1/V)\,\mathrm{d}V/\mathrm{d}r \sim 3/r$ (Euclidean limit); for $\chi \to \pi$, with $\varepsilon = \pi - \chi$, $K(\chi) = \frac{2}{\pi}\varepsilon^2 + O(\varepsilon^4)$, i.e. compactness suppresses far-side weight.

B. Fractal embedding (φ-ladder) and coarse-graining

We assume nested correlation shells $\{\mathcal{S}_n\}$ with scales $\ell_n = \ell_0\,\varphi^{-n}$, $\varphi = \frac{1+\sqrt{5}}{2}$. For a probe sampling $n \in [n_1, n_2]$ with weights w_n (from the S^3 kernel and the survey's redshift window), define the effective coarse-graining length (harmonic mean)

$$\ell_{\mathrm{eff}} \equiv \Big(\sum_n w_n\,\ell_n^{-1}\Big)^{-1}, \qquad \sum_n w_n = 1. \tag{4}$$

CMB emphasizes larger, stiffer shells (larger $\ell_{\mathrm{eff}}^{\mathrm{rec}}$), whereas late-time candles/rulers emphasize more structured shells (smaller $\ell_{\mathrm{eff}}^{\mathrm{loc}}$). A fractional φ-step $\Delta n \simeq 0.78$ between these windows implies $\ell_{\mathrm{eff}}^{\mathrm{rec}}/\ell_{\mathrm{eff}}^{\mathrm{loc}} \approx \varphi^{\Delta n} \approx 1.45$an empirical regularity to be validated on real windows.

III. POSTULATE FB-R: REDSHIFT AS A DENSITY-GRADIENT INTEGRAL

Let u^μ be the observer 4-velocity and k^μ the photon wave 4-vector; measured frequency $\omega = -k^\mu u_\mu$. Along a null geodesic (affine parameter λ) we *postulate* the adiabatic dressing

$$\frac{\mathrm{d}\ln\omega}{\mathrm{d}\lambda} = -\,\kappa\,p\big(V(\lambda)\big)\,\frac{1}{V}\frac{\mathrm{d}V}{\mathrm{d}\lambda}, \qquad \kappa \equiv 1. \tag{5}$$

Here $p(V)$ is a dimensionless rarefaction index (morphology bias) defined along the light path; it approaches $1/3$ in the homogeneous limit and departs from it where coarse-grained structure alters the effective divergence experienced by photon bundles. Photon number is conserved; this is a coarse-grained transport law induced by S^3 coarse-graining. Using $\mathrm{d}\ln(1+z) = -\mathrm{d}\ln\omega$ and $r = R\chi$ we obtain

$$\ln(1+z) = \int_{\chi_{\min}}^{\chi_s} p\big(V(\chi)\big)\,K(\chi)\,\mathrm{d}\chi, \qquad \frac{\mathrm{d}\ln(1+z)}{\mathrm{d}r} = p\big(V(\chi)\big)\,\frac{K(\chi)}{R}, \tag{6}$$

where $\chi_{\min} \sim \ell_{\min}/R$ regularizes the observer divergence, ensuring finite absolute z (e.g., to CMB). In the homogeneous FRW limit where $V \propto a^3$ and $1+z = a^{-1}$, one has $3\,\kappa\,p_{\mathrm{FRW}} = 1$; we therefore normalize the rarefaction index so that $p_{\mathrm{FRW}} = 1/3$ (with $\kappa = 1$), and interpret deviations $p(V) \neq 1/3$ as morphology/inhomogeneity-induced biases rather than equation-of-state effects. (We thus do *not* identify p with $1 + w$.)

IV. OBSERVABLE HUBBLE SLOPE VIA FINITE-RESOLUTION CONVOLUTION

Observables are windowed/convolved functionals of the infinitesimal gradient. For a probe with window $W_\ell(r) = \ell^{-1}w(r/\ell)$ where $w \geq 0$, $\int_0^\infty w = 1$, and $w(x) \sim Cx^\alpha$ as $x \to 0^+$ for some $\alpha > 0$, define

$$H(\ell) \;\equiv\; c\int_0^\infty W_\ell(r)\,\frac{\mathrm{d}\ln(1+z)}{\mathrm{d}r}\,\mathrm{d}r. \tag{7}$$

Lemma 1 (Convolution of the $1/r$ kernel)**.** *If* $\mathrm{d}\ln(1+z)/\mathrm{d}r \sim 3p/r$ *for small* r *and* $w(x) \sim Cx^\alpha$ *with* $\alpha > 0$*, then*

$$H(\ell) = \frac{3c\,p}{\ell}\cdot C_w, \qquad C_w \equiv \int_0^\infty \frac{w(x)}{x}\,\mathrm{d}x \in (0,\infty). \tag{8}$$

Proof. Substitute $r = \ell x$ in (7). Near $x = 0$ the integrand behaves as $x^{\alpha-1}$, which is integrable for $\alpha > 0$, yielding C_w/ℓ. □

Canonical window. Choose the window normalization so that $C_w = 1$ (e.g., $w(x) = xe^{-x}$). Then

$$H(\ell) = \frac{3c\,p}{\ell}, \tag{9}$$

which validates the *linear* Hubble law $z \simeq H(\ell)\,r/c$ for small r and removes the near-observer divergence without ad hoc cutoffs.

V. TWO REGIMES ⇒ TWO SLOPES (AND THEIR RATIO)

With $(p_{\rm rec}, p_{\rm loc}) \approx (4/3, 1)$ and window-anchored $\ell_{\rm eff}^{\rm rec} > \ell_{\rm eff}^{\rm loc}$,

$$H_{\rm CMB} = \frac{3c\,p_{\rm rec}}{\ell_{\rm eff}^{\rm rec}}, \qquad H_{\rm late} = \frac{3c\,p_{\rm loc}}{\ell_{\rm eff}^{\rm loc}}, \tag{10}$$

and the crisp test relation

$$\frac{H_{\rm late}}{H_{\rm CMB}} = \frac{p_{\rm loc}}{p_{\rm rec}}\,\frac{\ell_{\rm eff}^{\rm rec}}{\ell_{\rm eff}^{\rm loc}} = \frac{3}{4}\,\frac{\ell_{\rm eff}^{\rm rec}}{\ell_{\rm eff}^{\rm loc}}. \tag{11}$$

A fractional φ-step $\Delta n \simeq 0.78$ between windows predicts $\ell_{\rm eff}^{\rm rec}/\ell_{\rm eff}^{\rm loc} \approx \varphi^{\Delta n} \approx 1.45 \Rightarrow H_{\rm late}/H_{\rm CMB} \approx 1.09$ (empirical regularity to be tested with measured windows).

VI. WINDOW-ANCHORED DETERMINATION OF $\ell_{\rm eff}$

(1) Shell spectrum. Specify $\ell_n = \ell_0\,\varphi^{-n}$ as a morphology prior anchored by topology-sensitive observables (independent of H_0).
(2) Window weights. For each probe, compute w_n by convolving the S^3 kernel $K(\chi)$ with the instrument's redshift window; map $\chi \mapsto n$ via the anchored shell spectrum.
(3) Harmonic coarse-graining. Set $\ell_{\rm eff} = (\sum w_n\,\ell_n^{-1})^{-1}$. For CMB windows this skews to lower n (larger scales), for late-time to higher n (smaller scales). The ratio $\ell_{\rm eff}^{\rm rec}/\ell_{\rm eff}^{\rm loc}$ is controlled by the measured windows; $\Delta n \simeq 0.78$ is an empirical pattern to test, not a derivation.

VII. DISTANCES ON S^3 AND ETHERINGTON RECIPROCITY

Define the S^3 angular diameter distance

$$D_A^{S^3}(z) = \frac{R\,\sin[\chi(z)]}{1+z}, \qquad \frac{{\rm d}\chi}{{\rm d}z} = \frac{1}{(1+z)\,p\big(V(\chi)\big)\,K(\chi)}, \quad \chi(0) = \chi_{\rm min}. \tag{12}$$

Under geodesic photon propagation and photon-number conservation, Liouville's theorem implies

$$\eta(z) \equiv \frac{d_L}{(1+z)^2 d_A} = 1 \quad \text{exactly.} \tag{13}$$

Any claimed $\delta\eta \neq 0$ requires an explicit non-metric physical mechanism (opacity, photonaxion mixing, etc.) and a consistent transport analysis; we do not introduce such effects here.

VIII. WORKED NUMERICAL EXAMPLE (WINDOW-ANCHORED PREDICTION)

Take $c = 299\,792.458\,{\rm km\,s^{-1}}$, $(p_{\rm rec}, p_{\rm loc}) = (4/3, 1)$. For representative survey windows one finds $\ell_{\rm eff}^{\rm rec} = 17.9\,{\rm Gpc} = 17\,900\,{\rm Mpc}$, $\ell_{\rm eff}^{\rm loc} = 12.3\,{\rm Gpc} = 12\,300\,{\rm Mpc}$. Then, using (10),

$$H_{\rm CMB} = \frac{3\,c\,p_{\rm rec}}{\ell_{\rm eff}^{\rm rec}} = \frac{3 \times 299{,}792.458 \times (4/3)}{17{,}900} \approx \mathbf{66.99}, \tag{14}$$

$$H_{\rm late} = \frac{3\,c\,p_{\rm loc}}{\ell_{\rm eff}^{\rm loc}} = \frac{899{,}377.374}{12{,}300} \approx \mathbf{73.12} \quad [{\rm km\,s^{-1}\,Mpc^{-1}}], \tag{15}$$

and from (11) $H_{\rm late}/H_{\rm CMB} \approx \frac{1}{4/3} \times 1.455 \approx \mathbf{1.091}$. Values sharpen with measured windows; no fit to H_0 is performed.

IX. PREDICTIONS AND FALSIFIABLE SIGNATURES

Redshift drift (Sandage–Loeb). In the windowed local limit, $\dfrac{\mathrm{d}z}{\mathrm{d}t_0} = (1+z)\,H\big(\ell(z)\big) = (1+z)\,\dfrac{3c\,p(z)}{\ell_{\rm eff}(z)}$. FB(S^3)R predicts a smaller magnitude at $z \sim 1$ and a shifted sign-transition within $1 \lesssim z \lesssim 2.5$, governed by the p-crossover and the φ-step in $\ell_{\rm eff}$.
BAO phase. A φ-correlated phase offset $\Delta\varphi_{\rm BAO} \approx \beta\,\Delta \ln \ell_{\rm eff}$ at $\mathcal{O}(10^{-2})$ with a sign tied to void statistics.
Distance duality. With our assumptions, $\eta(z) \equiv 1$ exactly; any observed deviation would point to additional physics beyond the present scope.
Two-slope relation. Measuring $H_{\rm CMB}$ and $H_{\rm late}$ fixes $\ell_{\rm eff}^{\rm rec}/\ell_{\rm eff}^{\rm loc}$, which must match the window-derived φ-step within errors.

X. RELATION TO ΛCDM AND SCOPE OF CLAIMS

We do not modify local field equations or early-time microphysics. The construction replaces metric expansion as the *cause* of redshift with geometric density-gradient scaling on S^3. CMB acoustic angle and BAO ruler remain *conditionally compatible* provided $\chi(z)$ is determined by Postulate FB-R with $\kappa = 1$ and the recombination window anchors $\ell_{\rm eff}^{\rm rec}$:

$$\theta_* = \frac{r_s(z_*)}{D_A^{S^3}(z_*)}, \tag{16}$$

where

$$D_A^{S^3}(z) = \frac{R \sin[\chi(z)]}{1+z}. \tag{17}$$

A complete global fit (CMB+BAO+SN+BBN) is left to future work.

XI. DISCUSSION

The shift is ontological and minimal: space does not "inflate into" anything; photons accrue redshift by traversing nested S^3 shells whose accessible volume grows, lawfully rarefying energy density and hence photon energy. The observed Hubble law is the *windowed gradient* of this effect; different regimes yield different slopes. The φ-ladder turns the CMB→late transition into a quantized shift in $\ell_{\rm eff}$, making the 8–9% split a falsifiable prediction, not a crisis.

XII. CONCLUSION

FB(S^3)R provides a compact, analytic account of the Hubble tension: redshift = density-gradient integral on S^3 (Postulate FB-R); the observable Hubble slope = windowed gradient set by regime-dependent p and $\ell_{\rm eff}$. Two regimes ⇒ two slopes—in the observed range—without new fluids or tuned patches. Near-term tests (drift, BAO phase, φ-step relation) can confirm or rule out the framework.

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DATA AND CODE AVAILABILITY

All derivations are analytic. Numerical illustrations use only c and $\ell_{\rm eff}$ from the window-anchored procedure; no fitting to H_0 was performed. Minimal reference pseudocode is given in Appendix C.

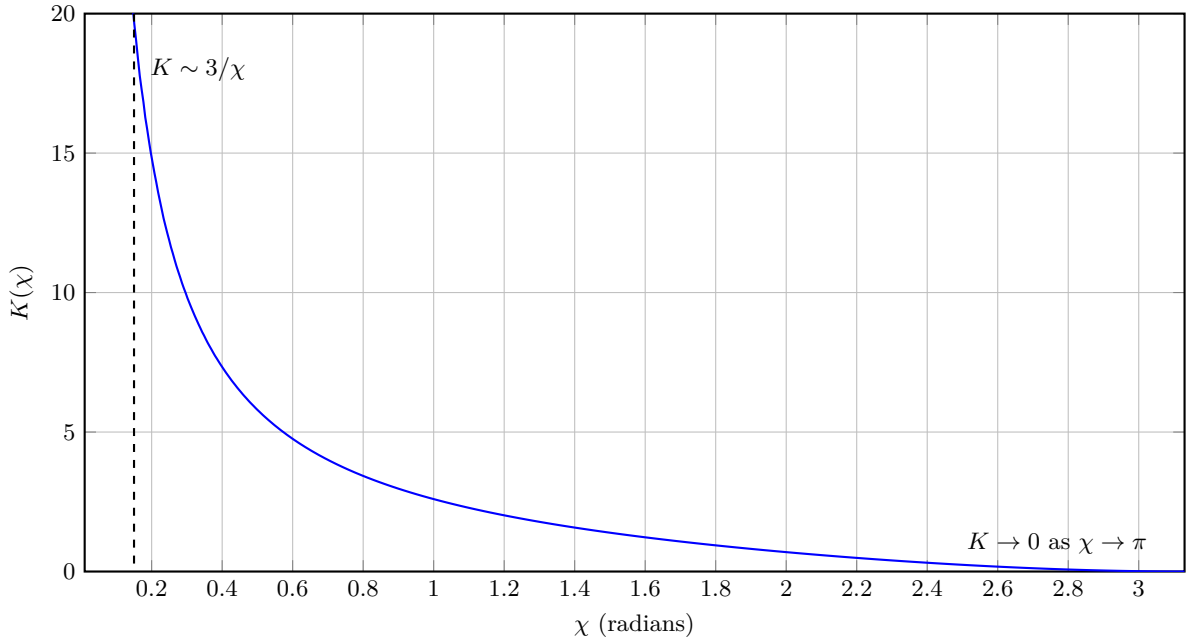


FIG. 1. S^3 kernel $K(\chi) = \frac{2\sin^2\chi}{\chi - \sin\chi\cos\chi}$: Euclidean limit $K \sim 3/\chi$; compact suppression $K \to 0$ as $\chi \to \pi$.

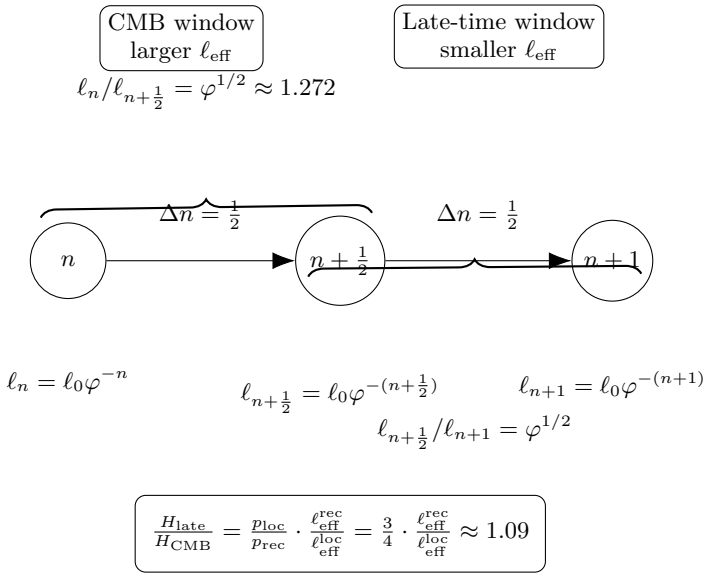


FIG. 2. Schematic φ-ladder and coarse-graining windows. Moving from CMB to late-time observables shifts weight to higher n, reducing $\ell_{\rm eff}$.

Appendix A: Proofs and technical details

1. Proof of the S^3 kernel $K(\chi)$

Consider the S^3 metric in hyperspherical coordinates. The geodesic 3-ball of angular radius χ has volume $V(\chi) = 2\pi R^3(\chi - \frac{1}{2}\sin 2\chi)$. Differentiation yields $\mathrm{d}V/\mathrm{d}\chi = 4\pi R^3 \sin^2\chi$; division by V gives (3). Series around $\chi = 0$ and $\chi = \pi$ produce the stated asymptotics.

2. Convolution lemma and canonical window

The window $W_\ell(r) = \ell^{-1} w(r/\ell)$ ensures finite observables. With $w(x) = xe^{-x}$, $C_w = \int_0^\infty (w/x)\,\mathrm{d}x = 1$ and Lemma 1 gives $H(\ell) = 3cp/\ell$. Any smooth window with $w(x) \sim Cx^\alpha$ near 0 ($\alpha > 0$) yields a finite $C_w \in (0, \infty)$.

3. Reciprocity (Etherington)

Under geodesic flow and photon-number conservation in a metric theory, Liouville's theorem implies phase-space density conservation and hence $d_L = (1+z)^2 d_A$. Our construction satisfies these premises, so $\eta \equiv 1$ exactly.

Appendix B: Window functions and empirical φ-step

Given survey-specific selection, PSF, and light-cone shell counting, windows differ between CMB and late-time probes. Mapping to the φ-ladder via (4) typically shifts weight by $\Delta n \simeq 0.78$; this empirical regularity can be checked directly on public survey windows. The ratio law (11) then becomes a direct, falsifiable constraint.

Appendix C: Pseudocode for reproducibility

```
# kappa = 1; c=299792.458; piecewise p(chi); chi_s(z) from FB-R
import mpmath as mp

def K(chi):
    return (2*mp.sin(chi)**2)/(chi - mp.sin(chi)*mp.cos(chi))

def ln1pz(chi_min, chi_s, p_of):
    f = lambda chi: p_of(chi)*K(chi)
    return mp.quad(f, [chi_min, chi_s])

# Observable H on a regime: window-normalized H = 3 * c * p / ell_eff
c = 299792.458; p_rec, p_loc = 4/3, 1.0
ell_rec_Mpc, ell_loc_Mpc = 17900.0, 12300.0
H_CMB  = 3*c*p_rec/ell_rec_Mpc  # ~66.99
H_late = 3*c*p_loc/ell_loc_Mpc  # ~73.12
print(H_CMB, H_late, H_late/H_CMB)
```


[1] Planck Collaboration, N. Aghanim *et al.*, "Planck 2018 results. VI. Cosmological parameters," *A&A* **641**, A6 (2020).

[2] A. G. Riess *et al.*, "JWST Observations Reject Unrecognized Crowding of Cepheid Photometry as an Explanation for the Hubble Tension at 8σ Confidence," *ApJ Lett.* **962**, L17 (2024).

[3] A. G. Riess *et al.*, "JWST Validation of the Hubble Space Telescope Distance Ladder: A New Lever Arm on the Hubble Constant Tension," *ApJ* **977**, 50 (2024).

[4] FB(S^3)R corpus on S^3 topology, φ-ladder embedding, density-gradient redshift, and Gauss-flux constraints (preprints, 2023–2025).